

Daily Algebra Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Remainder & Factor Theorem · AM–GM & algebraic inequalities · Completing the square in two variables · Rational Root Theorem · Modular/divisibility arguments · Parametric families of solutions.

Section A Rapid Recognition

1. $f(x) = x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k .

Answer: $k = 6$

Method: Factor Theorem: if $(x - 2)$ is a factor then $f(2) = 0$. $8 - 12 + 2k - 8 = 0 \Rightarrow 2k = 12 \Rightarrow k = 6$. Never do polynomial long division when the factor theorem gives a one-line equation.

Investigate further: Remainder Theorem (full statement): when $p(x)$ is divided by $(x - a)$, the remainder is $p(a)$. Use this to find the remainder when $f(x) = x^3 - 6x^2 + 11x - 6$ is divided by $(x - 4)$. Then explore: if the remainder when $f(x)$ is divided by $(x - 1)$ is 3 and by $(x + 1)$ is -1 , find the remainder when divided by $(x^2 - 1)$. (Key: the remainder is now linear: $ax + b$. Set up two equations.)

2. Find the remainder when $p(x) = x^{100} - 1$ is divided by $(x - 1)$.

Answer: 0

Method: $p(1) = 1^{100} - 1 = 0$. So the remainder is 0 and $(x - 1)$ is in fact a factor. The Remainder Theorem makes this instantaneous — no division required.

Investigate further: Find the remainder when $x^{100} + x^{50} + 1$ is divided by $(x - 1)$, $(x + 1)$, and $(x^2 - 1)$. Then prove: $x^n - 1$ is always divisible by $(x - 1)$ for positive integers n . (Use the factorisation $x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \dots + 1)$, which you can verify by multiplying out.)

3. Factorise: $6x^2 - x - 12$.

Answer: $(2x - 3)(3x + 4)$

Method: Product = $6 \times (-12) = -72$. Seek pair summing to -1 : -9 and 8 . Split: $6x^2 - 9x + 8x - 12 = 3x(2x - 3) + 4(2x - 3) = (2x - 3)(3x + 4)$.

Investigate further: AC method / discriminant connection. A quadratic $ax^2 + bx + c$ factorises over $\mathbb{Z}$ if and only if its discriminant $b^2 - 4ac$ is a perfect square. Check: $(-1)^2 - 4(6)(-12) = 1 + 288 = 289 = 17^2$. ✓ This gives the roots directly: $x = \frac{1 \pm 17}{12}$, i.e. $x = \frac{3}{2}$ or $x = -\frac{4}{3}$. Investigate: for which integer values of c does $6x^2 - x + c$ factorise over $\mathbb{Z}$?

4. Simplify: $\frac{x^{2n} - 1}{x^n - 1}$.

Answer: $x^n + 1$

Method: Difference of two squares: $x^{2n} - 1 = (x^n - 1)(x^n + 1)$. Cancel $(x^n - 1)$. This is the $m = 2$ case of the general factorisation $x^{mn} - 1$.

Investigate further: Generalise: $\frac{x^{mn} - 1}{x^n - 1} = x^{n(m-1)} + x^{n(m-2)} + \cdots + x^n + 1$, a geometric series in x^n . Use this with $x = 10$, $n = 1$ to explain why $\underbrace{99 \cdots 9}_m$ is always divisible by 9. Then use $n = 2$, $m = 3$ to factorise $x^6 - 1$ fully over $\mathbb{Z}$.

5. Given $x > 0$, find the minimum value of $x + \frac{9}{x}$.

Answer: 6

Method: AM–GM: $x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 2 \cdot 3 = 6$. Equality when $x = \frac{9}{x}$, i.e. $x = 3$. Always suspect AM–GM when you see a sum of a quantity and its reciprocal (scaled).

Investigate further: Generalise: the minimum of $x + \frac{c}{x}$ for $x, c > 0$ is $2\sqrt{c}$, achieved at $x = \sqrt{c}$. Now find the minimum of $4x + \frac{9}{x}$ for $x > 0$, and in general of $ax + \frac{c}{x}$ for $a, c > 0$. (Answer: $2\sqrt{ac}$, at $x = \sqrt{c/a}$.)

6. Factorise over $\mathbb{Z}$: $x^4 - y^4$.

Answer: $(x^2 + y^2)(x + y)(x - y)$

Method: Apply difference of two squares twice: $x^4 - y^4 = (x^2 + y^2)(x^2 - y^2) = (x^2 + y^2)(x + y)(x - y)$. The factor $x^2 + y^2$ does not factorise further over $\mathbb{R}$.

Investigate further: Over $\mathbb{C}$, $x^2 + y^2 = (x + iy)(x - iy)$. So $x^4 - y^4 = (x + iy)(x - iy)(x + y)(x - y)$ over $\mathbb{C}$. Now factorise $x^4 - 1$ fully over $\mathbb{R}$, then over $\mathbb{C}$. Connect to the *roots of unity*: the four roots of $x^4 = 1$ are $1, -1, i, -i$, corresponding exactly to the four factors.

7. The roots of $2x^2 - 6x + 3 = 0$ are α and β . Find $\alpha^2 + \beta^2$.

Answer: 6

Method: Vieta: $\alpha + \beta = 3$, $\alpha\beta = \frac{3}{2}$. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 3 = 6$. Note: for a non-monic quadratic $ax^2 + bx + c = 0$, Vieta gives $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.

Investigate further: Find $\alpha^3 + \beta^3$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ using the same roots. For $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$, write it as $\frac{\alpha^2 + \beta^2}{\alpha^2\beta^2}$. Then investigate: what quadratic has roots α^2 and β^2 ? (Use the Vieta data for the new quadratic: sum = $\alpha^2 + \beta^2$, product = $\alpha^2\beta^2$.)

8. Simplify: $\frac{(n+1)! - n!}{(n-1)!}$.

Answer: n^2

Method: Factor: $(n+1)! - n! = n! \cdot n = n \cdot n!$. Then $\frac{n \cdot n!}{(n-1)!} = n \cdot \frac{n!}{(n-1)!} = n \cdot n = n^2$. Always

factor out the smaller factorial first.

Investigate further: Prove the identity $(n+1)! - n! - n \cdot (n-1)! = (n-1)! \cdot (n^2 - n)$ by factoring $(n-1)!$ throughout. Then investigate: for which n is $\frac{(2n)!}{(n!)^2}$ (the central binomial coefficient) divisible by $n+1$? (*Catalan number connection.*)

9. Show that $(a+b+c)^2 - (a^2 + b^2 + c^2) = 2(ab+bc+ca)$ and state $ab+bc+ca$ when $a+b+c=4$, $a^2+b^2+c^2=8$.

Answer: $ab+bc+ca=4$

Method: Expand $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$. Rearrange. Substituting: $16 - 8 = 2(ab+bc+ca) \Rightarrow ab+bc+ca=4$.

Investigate further: With $a+b+c=4$, $a^2+b^2+c^2=8$, $ab+bc+ca=4$: now use Newton's identity to find $a^3+b^3+c^3$ in terms of the elementary symmetric polynomials $e_1=4$, $e_2=4$, $e_3=abc$. Show that $a^3+b^3+c^3=3e_3+e_1(e_1^2-3e_2)=3abc+16$. Hence $a^3+b^3+c^3$ depends on abc — find specific triples (a,b,c) satisfying the constraints and verify.

10. Find the value(s) of k such that $x^2+kx+16$ is a perfect square.

Answer: $k=\pm 8$

Method: A perfect-square quadratic has discriminant zero: $k^2-64=0 \Rightarrow k=\pm 8$. Then $x^2+8x+16=(x+4)^2$ and $x^2-8x+16=(x-4)^2$.

Investigate further: Generalise: x^2+kx+c is a perfect square iff $k=\pm 2\sqrt{c}$. For c to allow integer k , we need c to be a perfect square itself. Now: for which integer values of m is x^2+mx+m a perfect square (i.e. has discriminant zero)? Find all such m .

Section B Manipulation Drills

1. Find all rational roots of $2x^3-3x^2-11x+6$ and factorise completely.

Answer: $(x-3)(x+2)(2x-1)$; rational roots: $x=3, -2, \frac{1}{2}$

Method: Rational Root Theorem: rational roots have the form $\pm \frac{p}{q}$ where $p \mid 6$ and $q \mid 2$. Candidates: $\pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{1}{2}, \pm \frac{3}{2}$. Test $x=3$: $54-27-33+6=0$. ✓ Divide out $(x-3)$ to get $2x^2+3x-2=(2x-1)(x+2)$.

Investigate further: Rational Root Theorem (formal statement): if $a_n x^n + \dots + a_0$ has a rational root p/q in lowest terms, then $p \mid a_0$ and $q \mid a_n$. Use this to show that x^3-2 has no rational roots, hence $\sqrt[3]{2}$ is irrational. Generalise: prove that $\sqrt[n]{k}$ is irrational whenever k is not a perfect n th power.

2. Simplify: $\frac{x^3+1}{x^2-x+1}$.

Answer: $x+1$

Method: Sum of cubes: $x^3+1=(x+1)(x^2-x+1)$. Cancel (x^2-x+1) . Check: $x^2-x+1 \neq 0$ for real x (discriminant < 0).

Investigate further: The factor x^2-x+1 is the *cyclotomic polynomial* $\Phi_6(x)$. Its roots are the

primitive 6th roots of unity: $e^{\pm i\pi/3}$. Show that $x^6 - 1 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$ by factorising step by step. This connects to the factorisation of $x^n - 1$ into cyclotomic polynomials, a key tool in number theory and algebra.

3. Given $a + b = p$ and $a^3 + b^3 = q$, express ab in terms of p and q .

Answer: $ab = \frac{p^3 - q}{3p}$

Method: $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)[(a + b)^2 - 3ab] = p(p^2 - 3ab) = q$. So $p^3 - 3p \cdot ab = q$, giving $ab = \frac{p^3 - q}{3p}$.

Investigate further: Given only $a + b = p$ and $a^3 + b^3 = q$, find $a^4 + b^4$ and $a^5 + b^5$ using Newton's recurrence $s_n = (a + b)s_{n-1} - ab \cdot s_{n-2}$. Then: if p and q are both integers, is ab necessarily rational? Always? Find an example where $p, q \in \mathbb{Z}$ but $a, b \notin \mathbb{Q}$.

4. Simplify: $\frac{x^2(x + 1) - x(x + 1)^2}{x(x + 1)}$.

Answer: -1

Method: Factor the numerator: $x(x + 1)[x - (x + 1)] = x(x + 1)(-1)$. Cancel $x(x + 1)$. Result: -1 . Always look for a common factor in the numerator before combining fractions.

Investigate further: This expression equals -1 for all $x \neq 0, -1$. What does this mean geometrically? The function $f(x) = \frac{x^2(x + 1) - x(x + 1)^2}{x(x + 1)}$ is a rational function that simplifies to the constant -1 with two holes. Sketch it and compare to $y = -1$. Then investigate: when does a rational function simplify to a polynomial? What algebraic condition must hold?

5. Factorise: $a^3 - a^2b - ab^2 + b^3$.

Answer: $(a - b)^2(a + b)$

Method: Group: $(a^3 + b^3) - (a^2b + ab^2) = (a + b)(a^2 - ab + b^2) - ab(a + b) = (a + b)(a^2 - 2ab + b^2) = (a + b)(a - b)^2$.

Investigate further: Verify using the substitution $b = a$: the expression becomes 0, confirming $(a - b)$ is a factor. Then check $b = -a$: again 0, confirming $(a + b)$ is a factor. Since the polynomial is degree 3 with two linear factors, the third must also be linear — and we can read it off. This substitution method ("*plug in values to find factors*") is the heart of the factor theorem. Use it to factorise $a^4 - b^4 - a^3b + ab^3$.

6. Use $\sum_{r=1}^n r = \frac{n(n+1)}{2}$ to find $\sum_{r=1}^n (2r - 1)$.

Answer: n^2

Method: $\sum_{r=1}^n (2r - 1) = 2 \cdot \frac{n(n+1)}{2} - n = n(n+1) - n = n^2$. The sum of the first n odd numbers is always a perfect square. Elegant!

Investigate further: Prove the same result geometrically: the n th odd number $2n - 1$ is the number of extra unit squares added to an $n \times n$ grid to produce an $(n + 1) \times (n + 1)$ grid (the L-shaped “gnomon”). Then find $\sum_{r=1}^n (2r)$ and $\sum_{r=1}^n r^2$ using the identity $(r + 1)^3 - r^3 = 3r^2 + 3r + 1$, telescoped.

7. Solve: $\frac{2x - 1}{x + 3} - \frac{x + 2}{x - 1} = 1$.

Answer: $x = -\frac{1}{5}$

Method: Common denominator $(x + 3)(x - 1)$: $(2x - 1)(x - 1) - (x + 2)(x + 3) = (x + 3)(x - 1)$. LHS: $(2x^2 - 3x + 1) - (x^2 + 5x + 6) = x^2 - 8x - 5$. RHS: $x^2 + 2x - 3$. So $x^2 - 8x - 5 = x^2 + 2x - 3 \Rightarrow -10x = 2 \Rightarrow x = -\frac{1}{5}$. Check in the original: LHS = $\frac{-7/5}{14/5} - \frac{9/5}{-6/5} = -\frac{1}{2} + \frac{3}{2} = 1$. ✓

Investigate further: Rational equations of this form always reduce to a polynomial equation after clearing denominators. Investigate: under what conditions does clearing denominators in $\frac{P(x)}{Q(x)} = \frac{R(x)}{S(x)}$ produce extraneous solutions? (*Extraneous solutions satisfy the cleared equation but make a denominator zero.*) Give an example of an equation that gains an extraneous solution when cleared.

8. Simplify: $\left(\frac{1}{a} - \frac{1}{b}\right) \div \left(\frac{1}{a^2} - \frac{1}{b^2}\right)$.

Answer: $\frac{ab}{a + b}$

Method: Numerator: $\frac{b - a}{ab}$. Denominator: $\frac{b^2 - a^2}{a^2b^2} = \frac{(b - a)(b + a)}{a^2b^2}$. Dividing: $\frac{b - a}{ab} \cdot \frac{a^2b^2}{(b - a)(b + a)} = \frac{ab}{a + b}$.

Investigate further: Note that $\frac{ab}{a + b} = \frac{1}{\frac{1}{a} + \frac{1}{b}}$, the *half-harmonic mean*. The harmonic mean of a and b is $H = \frac{2ab}{a + b}$. Prove the chain of inequalities $H \leq G \leq A$ where $G = \sqrt{ab}$ and $A = \frac{a + b}{2}$ for $a, b > 0$. (Use *AM-GM* twice: once to get $G \leq A$, once rearranged to get $H \leq G$.)

9. Show that $n^3 - n$ is divisible by 6 for all integers n .

Answer: Divisible by 6 for all $n \in \mathbb{Z}$.

Method: $n^3 - n = n(n - 1)(n + 1) = (n - 1)n(n + 1)$: the product of three consecutive integers. *Divisible by 2:* among any two consecutive integers, one is even. *Divisible by 3:* among any three consecutive integers, one is divisible by 3. Since $\gcd(2, 3) = 1$, the product is divisible by 6. □

Investigate further: Generalisation. Show that the product of any k consecutive integers is divisible by $k!$. This is equivalent to showing that $\binom{n}{k} = \frac{n!}{k!(n - k)!}$ is always an integer for $n \geq k$. (*Proof:* $\binom{n}{k}$ counts the number of k -element subsets of an n -element set, hence is an integer.) Apply: show $n^5 - n$ is always divisible by 30 by factoring $n^5 - n = n(n^4 - 1)$ further.

10. Roots of $x^2 - x - 1 = 0$ are α, β . Find $\alpha^3 + \beta^3$ and $\alpha^4 + \beta^4$.

Answer: $\alpha^3 + \beta^3 = 4$; $\alpha^4 + \beta^4 = 7$

Method: Vieta: $\alpha + \beta = 1$, $\alpha\beta = -1$. Newton recurrence $s_n = (\alpha + \beta)s_{n-1} - \alpha\beta \cdot s_{n-2} = s_{n-1} + s_{n-2}$. $s_1 = 1$, $s_2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 + 2 = 3$. $s_3 = s_2 + s_1 = 4$. $s_4 = s_3 + s_2 = 7$. This recurrence is the *Lucas sequence* — the roots are the golden ratio and its conjugate.

Investigate further: Golden ratio and Fibonacci. The roots of $x^2 - x - 1 = 0$ are $\phi = \frac{1+\sqrt{5}}{2}$ and $\hat{\phi} = \frac{1-\sqrt{5}}{2}$ (the golden ratio and its conjugate). The power sums $s_n = \phi^n + \hat{\phi}^n$ satisfy $s_n = s_{n-1} + s_{n-2}$ (Fibonacci-like). Compute s_0 through s_8 and compare to the Lucas numbers: 2, 1, 3, 4, 7, 11, 18, 29, 47. Prove that $\phi^n = F_n\phi + F_{n-1}$ where F_n is the n th Fibonacci number.

Section C Substitution & Structure

1. Solve: $9^x - 10 \cdot 3^x + 9 = 0$.

Answer: $x = 0$ and $x = 2$

Method: Let $u = 3^x$: $u^2 - 10u + 9 = 0 \Rightarrow (u-1)(u-9) = 0$. $u = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$. $u = 9 \Rightarrow 3^x = 9 \Rightarrow x = 2$.

Investigate further: Solve $4^x - 5 \cdot 2^x + 4 = 0$ and $25^x - 26 \cdot 5^x + 25 = 0$ by the same method. Then investigate: $2^{2x} - k \cdot 2^x + k - 1 = 0$. Show that $x = 0$ is always a solution regardless of k , and find the second solution in terms of k . What happens when $k = 2$?

2. Find the minimum value of $x^2 + y^2$ subject to $x + 2y = 5$.

Answer: 5

Method: Substitute $x = 5 - 2y$: minimise $(5-2y)^2 + y^2 = 5y^2 - 20y + 25$. Complete the square: $5(y-2)^2 + 5$. Minimum is **5** at $y = 2$, $x = 1$. Alternatively (elegant): by Cauchy-Schwarz, $(x+2y)^2 \leq (x^2+y^2)(1^2+2^2)$, so $x^2 + y^2 \geq \frac{25}{5} = 5$.

Investigate further: Cauchy-Schwarz inequality: $(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2)$. Prove it by expanding $(a_1b_2 - a_2b_1)^2 \geq 0$. Use it to find the minimum of $x^2 + y^2 + z^2$ subject to $x + y + 2z = 6$. Then connect to the geometric interpretation: $\sqrt{x^2 + y^2}$ is the distance from the origin to the line $x + 2y = 5$, and the minimum is the perpendicular distance.

3. Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ for all real a, b, c .

Answer: Proof: see method.

Method: $2(a^2 + b^2 + c^2) - 2(ab + bc + ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$. Divide by 2. Equality iff $a = b = c$. $\square$ This is the most important algebraic inequality after AM-GM.

Investigate further: The expression $(a-b)^2 + (b-c)^2 + (c-a)^2$ is a *sum of squares (SOS) decomposition*, which proves non-negativity constructively. Now prove: $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2$ using the same idea (replace a, b, c with a^2, b^2, c^2). Then investigate: is $a^2 + b^2 + c^2 \geq ab + bc + ca$ still true if a, b, c are complex numbers? (No — consider what ' $\geq$ ' means for complex numbers.)

4. Solve: $\sqrt{x+5} + \sqrt{x-3} = 4$.

Answer: $x = 4$

Method: Isolate one radical: $\sqrt{x+5} = 4 - \sqrt{x-3}$. Square: $x+5 = 16 - 8\sqrt{x-3} + (x-3) \Rightarrow 8\sqrt{x-3} = 8 \Rightarrow x-3 = 1 \Rightarrow x = 4$. Check: $\sqrt{9} + \sqrt{1} = 3 + 1 = 4$. $\checkmark$

Investigate further: The conjugate trick: instead of isolating, multiply both sides by $(\sqrt{x+5} - \sqrt{x-3})$. LHS becomes $(x+5) - (x-3) = 8$, giving $\sqrt{x+5} - \sqrt{x-3} = 2$. Combined with the original equation: $2\sqrt{x+5} = 6 \Rightarrow x+5 = 9 \Rightarrow x = 4$. This conjugate approach is faster and avoids squaring. Use it to solve $\sqrt{x+4} - \sqrt{x-4} = \frac{8}{5}$.

5. Find the minimum of $x^2 + 4x + y^2 - 6y + 13$, and the (x, y) at which it occurs.

Answer: Minimum is 0, at $(x, y) = (-2, 3)$

Method: Complete the square in each variable separately: $(x+2)^2 - 4 + (y-3)^2 - 9 + 13 = (x+2)^2 + (y-3)^2 \geq 0$. Minimum is 0 when $x = -2, y = 3$.

Investigate further: Note $(x+2)^2 + (y-3)^2 = 0$ means the expression equals the square of the distance from $(-2, 3)$ to (x, y) . So the minimum is the point closest to the origin — but the expression vanishes at $(-2, 3)$, not at the origin. Now find the minimum of $x^2 + y^2 + z^2 - 2x + 4y - 6z + 14$. Is the minimum always achievable for quadratics in n variables? (Yes, iff the quadratic form is positive definite — a key idea in multivariable optimisation.)

6. If $x + \frac{1}{x} = 5$, find $\frac{x^4 + 1}{x^2}$.

Answer: 23

Method: $\frac{x^4 + 1}{x^2} = x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = 25 - 2 = 23$.

Investigate further: Extend: find $\frac{x^6 + 1}{x^3}$ and $\frac{x^8 + 1}{x^4}$ using the chain: $s_n = \left(x + \frac{1}{x}\right) s_{n-1} - s_{n-2}$ where $s_k = x^k + \frac{1}{x^k}$. This is the same Newton recurrence as before, applied to x and $\frac{1}{x}$ (whose sum is 5 and product is 1). Compute s_1 through s_5 .

7. Solve over $\mathbb{R}$: $x^4 + 4x^2 - 5 = 0$.

Answer: $x = \pm 1$

Method: Let $u = x^2$: $(u+5)(u-1) = 0$. $u = -5$ gives no real solutions. $u = 1 \Rightarrow x = \pm 1$.

Investigate further: For which values of k does $x^4 + 4x^2 + k = 0$ have: (i) no real solutions, (ii) exactly two real solutions, (iii) four real solutions? (Set $u = x^2$: the quadratic $u^2 + 4u + k = 0$ has roots that must be positive.) Sketch the curve $y = x^4 + 4x^2$ and read off the answers geometrically.

8. Given $a, b > 0$ with $a + b = 1$, find the minimum of $\frac{1}{a} + \frac{1}{b}$.

Answer: 4

Method: $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{1}{ab}$. By AM–GM, $ab \leq \left(\frac{a+b}{2}\right)^2 = \frac{1}{4}$, so $\frac{1}{ab} \geq 4$. Equality when $a = b = \frac{1}{2}$.

Investigate further: Reprove the result via Cauchy–Schwarz: $\left(\frac{1}{a} + \frac{1}{b}\right)(a+b) \geq (1+1)^2 = 4$, and $a+b=1$. Generalise: for $a_1 + a_2 + \dots + a_n = 1$ with all $a_i > 0$, find the minimum of $\sum_{i=1}^n \frac{1}{a_i}$ using the AM–HM inequality. (Answer: n^2 .)

Section D Challenge

(TMUA / SMC difficulty)

1. Find all real solutions to $x^4 + 2x^3 - 2x - 1 = 0$.

Answer: $x = 1$ and $x = -1$ ($x = -1$ has multiplicity 3)

Method: Group into two differences: $x^4 + 2x^3 - 2x - 1 = (x^4 - 1) + 2x(x^2 - 1)$. Both terms share the factor $x^2 - 1$: $(x^2 - 1)(x^2 + 1) + 2x(x^2 - 1) = (x^2 - 1)(x^2 + 2x + 1) = (x - 1)(x + 1)(x + 1)^2 = (x - 1)(x + 1)^3$. The only real solutions are $x = 1$ and $x = -1$.

Investigate further: The factorisation $x^4 + 2x^3 - 2x - 1 = (x - 1)(x + 1)^3$ shows $x = -1$ is a root of multiplicity 3. Verify by computing $f(-1)$, $f'(-1)$, $f''(-1)$ and checking they are all zero. Geometrically, a root of multiplicity 3 means the curve $y = f(x)$ crosses the x -axis and has an inflection point there simultaneously. Explore: what does multiplicity 4 look like? Sketch $y = (x - a)^4$ near $x = a$.

2. Prove that $n^2 + n + 41$ is not always prime by finding the smallest counterexample, and explain why it is prime for $n = 0, 1, \dots, 39$.

Answer: Smallest counterexample: $n = 40$, giving $40^2 + 40 + 41 = 41^2 = 1681 = 41 \times 41$.

Method: $n = 40$: $1600 + 40 + 41 = 1681 = 41^2$. Composite. $\square$ For $0 \leq n \leq 39$: one approach is to observe that $f(n) = n^2 + n + 41$ — any factor d of $f(n)$ with $d \leq 40$ would divide $f(n) - f(0) = n(n + 1)$ and also 41, forcing $d = 1$ or $d = 41$. Since $f(n) < 41^2$ for $n < 40$, the factor 41 cannot appear.

Investigate further: Euler's lucky numbers. The polynomial $n^2 + n + p$ is prime for $n = 0, 1, \dots, p - 2$ if and only if p is one of Euler's lucky numbers: 2, 3, 5, 11, 17, 41. This connects to the theory of *imaginary quadratic fields* with class number 1. Verify that $n^2 + n + 17$ is prime for $n = 0, \dots, 15$ and find the first counterexample. Then show directly that $n^2 + n + 41 \equiv 0 \pmod{41}$ iff $41 \mid n$ or $41 \mid (n + 1)$.

3. Let $a, b, c > 0$ with $abc = 1$. Prove that $a + b + c \geq 3$.

Answer: Proof via AM–GM: see method.

Method: By AM–GM on three positive numbers: $\frac{a + b + c}{3} \geq \sqrt[3]{abc} = \sqrt[3]{1} = 1$. Therefore $a + b + c \geq 3$. Equality iff $a = b = c = 1$. $\square$ This is the AM–GM inequality in its three-variable form, with the constraint doing all the work.

Investigate further: AM–GM for n variables: $\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}$ for positive reals, with equality iff all are equal. Use it to prove: for $a, b, c > 0$ with $a + b + c = 3$, we have $abc \leq 1$. (*This is the 'flipped' version of today's problem.*) Then prove: $a^2 + b^2 + c^2 \geq \frac{(a + b + c)^2}{3}$ (power-mean inequality) using $3(a^2 + b^2 + c^2) \geq (a + b + c)^2$ via the Cauchy–Schwarz inequality.

4. For positive reals a, b , prove $\frac{a}{b} + \frac{b}{a} \geq 2$. Hence prove AM–GM: $\frac{a + b}{2} \geq \sqrt{ab}$.

Answer: Proof: see method.

Method: *Part 1:* Let $t = \frac{a}{b} > 0$. Then $t + \frac{1}{t} - 2 = \frac{t^2 - 2t + 1}{t} = \frac{(t - 1)^2}{t} \geq 0$. So $\frac{a}{b} + \frac{b}{a} \geq 2$, equality iff $a = b$. $\square$

Part 2: Apply Part 1 to the positive reals $\sqrt{a}$ and $\sqrt{b}$: $\frac{\sqrt{a}}{\sqrt{b}} + \frac{\sqrt{b}}{\sqrt{a}} \geq 2 \Rightarrow \frac{a + b}{\sqrt{ab}} \geq 2 \Rightarrow \frac{a + b}{2} \geq \sqrt{ab}$.

□

Investigate further: These two proofs show that *every algebraic inequality ultimately rests on the fact that squares are non-negative*. Find at least three distinct proofs of AM–GM for two variables. Then prove the *weighted AM–GM*: for $\lambda \in (0, 1)$, $\lambda a + (1 - \lambda)b \geq a^\lambda b^{1-\lambda}$. (*This generalises AM–GM and is fundamental in convex analysis.*)

5. Find the number of integers n with $1 \leq n \leq 1000$ such that $(n + 3) \mid (n^2 - 1)$.

Answer: 2 ($n = 1$ and $n = 5$)

Method: $n^2 - 1 = (n + 3)(n - 3) + 8$, so $(n + 3) \mid (n^2 - 1)$ iff $(n + 3) \mid 8$. Since $n \geq 1$, we need $n + 3 \geq 4$, and the only divisors of 8 that are ≥ 4 are 4 and 8, giving $n = 1$ and $n = 5$. Exactly **2** values in range.

Investigate further: Generalise: find all $n \geq 1$ such that $(n + k) \mid (n^2 - 1)$ for fixed k . Write $n^2 - 1 = (n + k)(n - k) + (k^2 - 1)$, so the condition becomes $(n + k) \mid (k^2 - 1)$. This is a standard technique: *reduce the divisibility to a condition on a constant*. Use it to find all $n \geq 1$ with $(n + 2) \mid (n^2 + 1)$.

Top 5 Patterns Today

- Remainder & Factor Theorem.** To find the remainder of $f(x) \div (x - a)$: evaluate $f(a)$. To show $(x - a)$ is a factor: show $f(a) = 0$. Instantaneous — no long division.
- AM–GM and the non-negativity of squares.** Every algebraic inequality ultimately reduces to $(x - y)^2 \geq 0$ or a sum-of-squares. AM–GM ($\frac{a+b}{2} \geq \sqrt{ab}$) is the compressed form of this for products.
- Completing the square in two (or more) variables.** $(x + h)^2 + (y + k)^2$ is the canonical form for optimisation. The minimum is the squared distance from a point to the origin after shifting.
- Rational Root Theorem.** For $a_n x^n + \dots + a_0$, rational roots have the form p/q with $p \mid a_0$, $q \mid a_n$. Combined with the Factor Theorem: test candidates, divide out, repeat.
- Divisibility via algebraic identity.** To show $(n + k) \mid f(n)$: write $f(n) = Q(n) \cdot (n + k) + R$ where R is a constant. Then $(n + k) \mid f(n)$ iff $(n + k) \mid R$ — reducing an infinite problem to a finite one.

Common Traps to Avoid

- Applying AM–GM to negative numbers.** $\frac{a+b}{2} \geq \sqrt{ab}$ requires $a, b > 0$. If $a < 0$, the square root $\sqrt{ab}$ may not even be real. Always check the sign condition before applying any mean inequality.
- Introducing extraneous solutions when squaring.** Squaring $\sqrt{A} = B$ requires $B \geq 0$ *first*. Always substitute back into the original equation to check.
- Forgetting that the Rational Root Theorem gives candidates, not roots.** Every candidate must be tested. Not every p/q is a root — the theorem only rules out impossible rational roots, it does not guarantee any exist.
- Confusing divisibility direction.** $(n + k) \mid (n^2 - 1)$ does *not* mean $(n^2 - 1) \mid (n + k)$. Write out the algebra explicitly: $a \mid b$ means $b = ka$ for some integer k .

5. **Treating multiplicity- k roots as k distinct solutions.** $x = -1$ is a root of $(x+1)^3(x-1) = 0$ with multiplicity 3 but it is *one* value of x . Count solutions, not multiplicities, unless the question asks.

“Mathematics is not about numbers, equations or algorithms: it is about understanding.”
— William Paul Thurston

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths